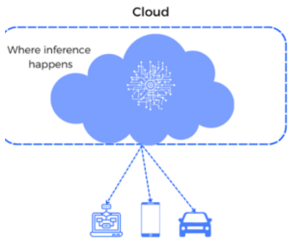
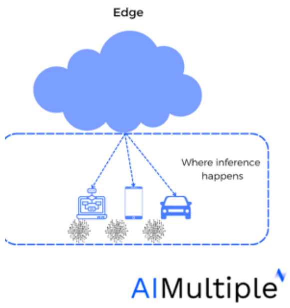
Edge vs Cloud Inference



Cloud Inference

☁ Data sent to cloud for processing
🧠 Supports complex & large AI models
✓ Centralized management & analytics
⚠ Introduces latency & network dependency



Edge Inference

📍 Models run on devices or gateways
⚡ Ultra-low latency & real-time response
📉 Reduced bandwidth usage
✓ Reliable even with limited connectivity

Hybrid Approach (Most Common)

⚡ Real-time decisions at the edge
☁ Deep analysis & model training in the cloud

Choosing the Right Approach

⏱ Latency requirements
💰 Cost & bandwidth considerations
🌐 Connectivity & operational constraints

Balanced edge–cloud intelligence delivers optimal performance

Key Comparison at a Glance

Feature	Edge Inference	Cloud Inference
Latency	Ultra-low (sub-50 ms)	Higher (100–500+ ms)
Primary Benefit	Real-time action, offline capability	Massive compute, scalability
Data Privacy	High (data stays local)	Moderate (data travels to server)
Compute Power	Limited by device hardware	Near-limitless
Best For	IoT, robotics, critical safety	Complex analytics, training

Use edge for speed & autonomy, cloud for scale & deep intelligence

Practical Implementation Strategies: Edge + Cloud AI

⚡ Prioritize Edge for Reflexive Actions

- ✓ Millisecond decision-making
  ✓ Operates during internet outages
  ✓ Protects sensitive personal data

☁ Leaverage Cloud for Deep Intelligence

- ✓ Run large-scale analytics & behavioural models
  ✓ Train & update AI models centrally
  ✓ Aggregate insights across devices

🔄 Hybrid Continuous Learning Loop

- 📍 Edge → real-time inference & actions
  ☁ Cloud → insight aggregation & model training
  🔧 Optimized models redeployed to edge

💰 Cost Management Considerations

- 📉 Edge reduces bandwidth & storage costs long-term
  💸 Cloud lowers entry cost but scales with usage
  ⚖ Balance CAPEX vs OPEX for sustainability

Combine edge speed with cloud intelligence for optimal AIoT performance